

TML2026: MODEL STEALING: STOLEN MODEL DETECTION

TASK 2 — CMS TEAM ID: TEAM_012

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1 GITHUB LINK

https://github.com/madhvanb/TML_Assignment.git

2 INTRODUCTION

Model stealing is a problem because in order to train a strong model we require a large amount of data, compute, time and expertise. If someone is able to then steal this model they can reuse the original training effort done without paying the same cost of resources. Even earlier work has shown models being extracted through APIs, which makes it a real problem in today's age. (1).

In this assignment, we worked from the perspective of a model owner trying to detect whether any suspect models were stolen from a given target model. The target model was a CIFAR-style ResNet-18 classifier trained on a subset of CIFAR-100, and the task was to assign a continuous stealing-confidence score to each of the 360 suspect models. A higher score means that the suspect model is more likely to be copied from, or derived from, the target model.

This is not a simple binary classification problem because stolen models may not always appear as exact copies. Some may be fine-tuned, post-processed, distilled, or modified in ways that reduce direct weight-level similarity. Because of this, we used a multi-signal approach instead of relying on only one test. Our final method combines weight similarity, behavioral agreement, representation similarity via CKA (2), and training-set memorization via Dataset Inference with a small complementary signal from boundary-proximal probes (5).

3 MAIN BODY

Our main idea was that if a suspect model is truly derived from the target, the relationship should appear in more than one form. We first compared the target model and each suspect model directly in weight space. For this purpose, we then used cosine similarity, L2 similarity, exact copy detection, per-layer cosine similarity, sorted-weight similarity and histogram based weight similarity. These metrics were helpful to determine which were direct copies and which were slightly modified ones. However, we found that weight similarity alone was not reliable enough, because independently trained models with the same architecture and a similar training setup can still appear close in weight space.

To capture functional similarity, we also compared the outputs of the target and suspect models on probe images. We used 1,000 images from the target training subset and 1,000 images that were excluded from that subset. For each suspect model, we measured top-1 agreement, top-5 overlap, logit cosine similarity, KL-based similarity, Spearman rank correlation over logits, and confidence-weighted agreement. These behavioral metrics helped us identify models that made similar predictions to the target, even when their weights were not identical.

The strongest signal in our approach was representation similarity using linear Centered Kernel Alignment, or CKA. CKA compares internal activation patterns between two models and is commonly used to measure similarity between neural network representations (2). We extracted ac-

tivations from the four main residual stages of the ResNet-18 model, from `layer1` to `layer4`. ResNet-style architectures are suitable for this kind of layer-wise comparison because they provide clear intermediate feature blocks (3). After applying global average pooling and centering the activations, we computed CKA between the target model and each suspect model. We then averaged the layer-wise CKA values into one representation-similarity score. This was useful because stolen or fine-tuned models may still learn similar internal features, even when their final predictions or raw parameters have changed. Representation-level comparison is also useful for model reuse detection, as discussed in ModelDiff (4).

We also used a Dataset Inference style signal to check whether a suspect model preserved the target model’s training-set behavior. Dataset Inference is based on the idea that a model can reveal information about whether certain data was part of its training process through differences in loss or behavior on selected examples (5). In our setting, the intuition was that a model derived from the target may perform unusually well on the exact images used to train the target, compared with images that were not part of the target training subset. For each suspect model, we computed the cross-entropy loss on in-set and out-set probe images and used the difference:

$$\text{DI gap} = \text{loss}_{\text{out}} - \text{loss}_{\text{in}}.$$

A larger positive gap suggests that the suspect model may have inherited the target model’s memorization pattern. We complemented the multi signal evaluation with a small set of boundary proximal probe which is a subset of CIFAR-100 test images with the highest predictive entropy under the target. On these inputs the target is least certain, and behavioural agreement on them is most discriminative between derived and independent models. The final stealing-confidence score combines the multi-signal evaluation (weight 0.70) with the boundary-proximal evaluation (weight 0.30), reflecting that CKA and Dataset Inference are the strongest individual signals while boundary-proximal behavior provides a complementary lift.

For the final stealing-confidence score, we normalized the different metrics and combined them using a weighted score. We assigned the highest weights to CKA (0.28) and the Dataset Inference gap (0.24), since these were the most informative signals in our experiments. Behavioral metrics such as top-1 agreement, KL similarity, and Spearman similarity were also included. Weight-based metrics served as supporting evidence with low weights. Final scores were normalized to [0, 1].

Our method identified five exact copies: models 71, 124, 138, 145, and 358. These models showed almost perfect agreement with the target across weights, predictions, and internal representations. Apart from the exact copies, CKA gave us a clear picture of the separation between the models that were derived from the ones that were independent. Around 80 suspect models that were tested had CKA values above 0.9, whereas most of the other models had low CKA values. ”The unified detector achieved $\text{TPR@5\%FPR} = 0.6111$.

4 CONCLUSION

Using just one metric for stolen model detection did not hold up well in our testing. Weight similarity fails when models get distilled, behavioral similarity gets fooled when copies are aggressively fine-tuned, and representations alone can miss quantized ones. CKA and Dataset Inference ended up being the most useful signals for us, and the bimodal CKA distribution basically gave us a clean split between the likely stolen suspects and the independent ones. Beyond the leaderboard, the practical side of model stealing is pretty serious. On the economic side, stolen models let attackers copy expensive training runs at almost no cost, which removes the incentive for companies to invest in ML research in the first place. On safety, a stolen model can bypass post-training alignment, meaning an attacker can extract the capabilities of a safety-tuned model and strip the safeguards off. On privacy, our Dataset Inference results show that stolen models inherit memorized training data, which creates risk for the people whose data was used. So reactive detection like ours is a necessary defense after the fact, but proactive measures like training-time watermarking are what would actually make stealing both detectable and not worth it economically.

REFERENCES

- [1] Florian Tramèr, Fan Zhang, Ari Juels, Michael K. Reiter, and Thomas Ristenpart. Stealing Machine Learning Models via Prediction APIs. In *USENIX Security Symposium*, 2016.

<https://arxiv.org/abs/1609.02943>

- [2] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of Neural Network Representations Revisited. In *International Conference on Machine Learning (ICML)*, 2019. <https://arxiv.org/abs/1905.00414>
- [3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. <https://arxiv.org/abs/1512.03385>
- [4] Yuanchun Nguyen et al. ModelDiff: Testing-Based DNN Similarity Comparison for Model Reuse Detection. In *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2022.
- [5] Adam Dziedzic, Mislav Dhawan, Nicolas Papernot, and Siddharth Garg. Dataset Inference for Self-Supervised Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. <https://arxiv.org/abs/2209.09024>